Requiem for Radio – New Dead Zones

README

Martin Marier

April 12, 2017

1 Description

The code in this repository is used in an outdoor sound sculpture titled *Requiem for Radio: New Dead Zones* by Amanda Dawn Christie. The piece is an hommage to the 13 shortwave emission towers that were in Sackville, NB. The piece is a scale model of the Sackville towers. The first version was set on a small hill by the boardwalk of the Halifax Harbour, next to the Sands at Salter, on October 15 2016.

When a visitor touches one of the sensors (one of the copper sponges or the copper pipe) on a tower, a sound is played back on a speaker positionned right next to it.

2 Technical overview

Each tower is equipped with a Wifi module (an ESP8266 on a Huzzah break out board, as distributed by Adafruit) and a capacitive sensing chip (MPR121, also on a break out board distributed by Adafruit). The MPR121 is an i2c device on which 12 electrodes can be attached. Each electrode is automatically calibrated (at boot time) to detect touch.

The SDA and SCL pins of the MPR121 are connected to the Huzzah board pins 4 and 5.

- i2c is on pins 4 and 5
  - SDA : 4
  - SCL : 5
- The IRQ is not used.

- UDP output port : 50101
- UDP input port (can be used to configure the MPR121) : 50202
- SSID : ndz
- Password : requiemforradionewdeadzones

The touch data and the raw data of each sensor is sent to a computer over Wifi (OSC messages over UDP) via a Wifi router. On the computer (a mac mini), a SuperCollider program receives the OSC messages and plays soundfiles on the appropriate speaker.

To get 13 analog audio outputs, two MOTU Ultralite AVB were used. The master card was the MOTU *adc* (Amanda's card) and the slave card was the MOTU *mm* (Martin's card). The cards were combined over AVB using a short cat-5 ethernet cable.

3 Router configuration

The router is already configured, but to make modifications, connect to it normally and, in a browser, enter `192.168.0.1` in the URL bar. To log into the router's configuration tool, use these:

User admin

Password Gq6ZaNRS5ICXw6UNqag8JiTKOGFfUf4T

The router is configured to assign a specific IP address to each tower using IP reservation tables.

Tower	IP address	Audio output
C	192.168.0.67	MOTU adc L
D	192.168.0.68	MOTU adc R
E	192.168.0.69	MOTU adc 1
F	192.168.0.70	MOTU adc 2
H	192.168.0.71	MOTU adc 3
I	192.168.0.73	MOTU adc 4
J	192.168.0.74	MOTU adc 5
K	192.168.0.75	MOTU adc 6
M	192.168.0.77	MOTU mm L
N	192.168.0.78	MOTU mm R
O	192.168.0.79	MOTU mm 1
P	192.168.0.80	MOTU mm 2
Q	192.168.0.81	MOTU mm 3

The last byte of the IP address corresponds to the ascii character of the tower (capitalized). This allows the SuperCollider program to identify each tower and to easily print the tower letter while debugging or testing.

4 Tower configuration and behavior

When a tower (an ESP8266) boots, it looks for the `ndz` SSID and tries to connect to it. If the connection fails (because the router was turned off or is out of range), the tower will have to be power cycled.

Once the connection is established, the tower starts sending UDP packets. These UDP packets contain OSC messages.

4.1 OSC messages

Two types of messages are sent:

/touch one 32-bit integer (only 13 bits are used). Each bit is the touch status of each electrode. Only four electrodes were used, but all 13 were enabled on the MPR121 and all the data was sent anyway.

/raw 13 32-bit integers (only 10 bits are used for each). These numbers are the raw values of the voltages (the inverse of the capacitance) measured for each electrode.

These messages are sent every 10 ms as long as the ESP8266 is powered. If the connection to the router is lost (if the router is rebooted, for example), the towers have to be rebooted so they can reconnect to it.

Port 50101 is used for OSC messages.

4.2 Copper pads

There are four copper pads on each tower. They should be connected to the pins 0, 1, 2 and 3 of the MPR121 chip (even if, at this point, it does not affect the behavior). The chip calibrates itself when it is powered on. This means that the pads shoud be connected to the chip before it is turned on. No matter what, power cycling the unit is the proper way to recalibrate it.

All the touch information is sent to SuperCollider, but the SC code does note differentiate between each copper pad. If any pad is touched, the audio (of a specific tower) is played back.

5 SuperCollider code

`main.scd` contains the code to execute to run RFR.NDZ.

TODO: find out and list the dependancies (quarks or other).

Executing the first line runs a test tone. 13 different test tones are output through the 13 first outputs of the sound card. The tones are all sine waves at -20 dBFS; the pitches are MIDI notes 60 through 72. This makes it easy to check if the speakers are wired properly.

RFR.NDZ is ran by executing the code between the parentheses.

5.1 Changing attack time

To be written.

5.2 AM synthesis and calibration

To enhance.

Uses the raw voltage of the pad connected to pin 0 to change the modulation frequency of a simple AM synth.

A calibration is required for this to work properly.

- Turn calibration on (on a specific tower or on all of them).
- Touch all the pads (firmly).
- Turn calibration off.

AM synthesis should only be turned on after calibration is done (and switched off).